A Short-Term Wind Power Scenario Generation Method Based on Conditional Diffusion Model

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***Abstract*— With the increasing integration of wind power into power systems, wind scenarios can pave the way for improved decision-making in the power system day-ahead scheduling to help manage the inherent uncertainty and variability of wind power. In this paper, a novel deep learning method that uses conditional diffusion model to address the challenges of wind scenario generation. The incorporation of the conditional term by considering the forecast errors distribution and the residual block with U-net structure was employed to handle complex non-linear relationships in temporal dependencies of short-term wind variations, enabling high-quality scenario generated outputs. A case study was conducted where four metrics was utilized to verify the effectiveness of proposed method, and a comparison with CGAN method was laid to demonstrate superiority of model performance.**

***Index Terms*—Conditional diffusion model, Scenario generation, Forecast errors distribution, Time series evaluation**

I. Introduction

Global renewable energy capacity has witnessed a rapid growth, increasing from 1566GW to 3372GW over the last decade[1]. China is one of the leading countries in development of renewable energy. With the proposal of carbon peaking and carbon neutrality target [2], renewable energy will gradually become the main power source of the new energy system. Wind power, as a technology form of renewable energy, plays a vital role in the transition towards a more sustainable and renewable energy future. In China, the installed capacity of wind power has reached 365GW, 14% of generation mix respectively by the end of 2022[3].

With the increasing integration of wind power into electricity systems, the inherent characteristics of wind power, such as intermittency, variability, and uncertainty, pose great challenges to day-ahead scheduling and the balance of electricity supply and demand in power system, so that it is important for system operators to know the expected wind power output in advance and make informed decisions. Therefore, the short-term wind power forecast, which provides a point-to-point time series curve describing deterministic wind power output value at each time point, is crucial for day-ahead scheduling.

As for a further step, multiple scenarios can be derived from forecasting data, which represent a spectrum of potential outcomes of wind power generation. Afterwards, scenario based stochastic optimization techniques and probabilistic modeling can be employed to quantify and manage the inherent variability and uncertainty of wind power.

In recent years, various methods have been developed in wind power scenarios generation. Commonly used approaches can be categorized into two types: statistical methods and deep-learning based methods.

Statistical methods establish a statistical model that describes the probability distribution of wind power output, then scenarios will be generated through specific sampling methods. Ref.[4] uses hidden Markov model to construct the correlated power time series of multiple wind farms. Ref.[5] utilizes quantile-based scenario tree structure to represent stochastic process of wind power output. Ref.[6] proposes a stochastic wind power model based on an Autoregressive Integrated Moving Average(ARIMA) process. Ref.[7] reproduces the statistical properties of the wind power by stochastic differential equations(SDE). On the other hand， the complexity of modeling on stochastic processes along with the significant impact of parameter selection on model accuracy severely hindered the applicability of statistical methods for scenario generation.

As an emerging approach, deep-learning based methods utilize deep neural networks to capture complex non-linear relationships and dependencies in the historical data, among which generative adversarial networks (GANs) are considered the state of the art in the field of scenario generation. Many variants of GANs such as WGAN[8], CGAN[9][10], Styled-GAN[11] have achieved promising results in implementing specific scenario generation tasks. However, consisted of a generator network and a discriminator network, which are trained simultaneously in a competitive manner, GANs may sometimes suffer from mode collapse and training instability.

Diffusion model[12], a novel framework for generative tasks, has achieved remarkable success in the field of audio synthesis[13] and image generation[14], but few

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research was focused on wind power scenarios generation. By comparison with deep learning based methods to date[15], diffusion model has a more structured and interpretable latent space, showcasing its versatility and effectiveness in generating high-quality outputs.

In this paper, a novel deep learning method is proposed for short-term wind power scenario generation based on conditional diffusion model. Firstly, we introduce the theory of diffusion model, especially the forward diffusion process and backward denoising process. Then, in regard with task of short-term wind power scenario generation, the distribution of forecast errors was analyzed, and a conditional diffusion model was designed to address this challenge. The RES-UNET network architecture was employed to capture temporal dependencies in the time series data and enhance training stability. In case study, four evaluation metrics were used, namely energy score, coverage rate, interval width, and autocorrelation coefficient to verify the effectiveness of the proposed approach. A comparison with the CGAN method was conducted, demonstrating the superiority of the proposed method.

The remainder of this paper is organized as follows: Section Ⅱ introduces the fundamental principles of diffusion models. Section Ⅲ describes short term scenarios generation based on conditional diffusion model. Section Ⅳ gives a case study, Conclusion is put in Section Ⅳ.

II. Diffusion Model

The core procedures of diffusion models are forward diffusion and backward denoising, which are shown in Fig.1. Concisely, forward diffusion is a process that iteratively adds noise vector into original data until the data distribution has been transformed to a "diffused" state. Backward denoising process aims at restoring the expected data by mirror operations of forward diffusion process. Both of procedures will be elaborated in detail below.

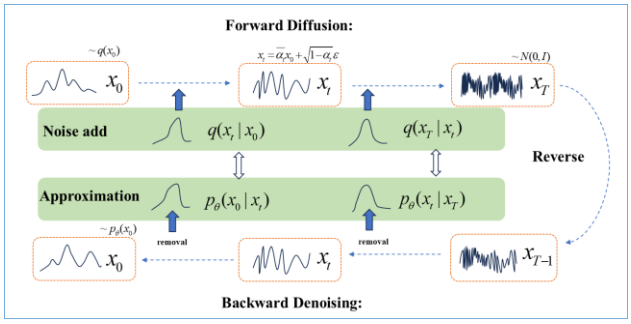

Fig. 1. Theory of diffusion model

A. Process of Forward Diffusion

Assume the original data x_0 has a distribution $x_0 \sim q(x_0)$, in each iteration, the gaussian noise is added sequentially into $x_0, x_1, \ldots, x_t, x_T$. The process can be parameterized as:

$$q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{1-\beta_t}\, x_{t-1}, \beta_t I)$$
$$\{\beta_t \in (0,1), \beta_0 < \beta_1 \ldots < \beta_{t-1} < \beta_t \ldots < \beta_T\} \tag{1}$$

Where T denotes total number of diffusion steps, β_t denotes a varying constant used to control the degree of noise at step t being added, and the value of β_t forms an increasing sequence to speed up diffusion process, $q(x_t \mid x_{t-1})$ denotes distribution of x_t under x_{t-1} follows a normal distribution with a mean of $\sqrt{1-\beta_t}\, x_{t-1}$ and a variance of β_t^2.

The probability distribution of current states x_t depends only on the last state x_{t-1}, known as the Markov property, implies that the entire diffusion process can be viewed as a Markov chain:

$$q(x_{1:T} \mid x_0) = \prod_{t=1}^{T} q(x_t \mid x_{t-1}) \tag{2}$$

According to equation (2), in process of forward diffusion, x_t can be inferred from x_0 and β_t:

$$x_t = \overline{\alpha_t} x_0 + \sqrt{1-\alpha_t}\,\varepsilon$$
$$\alpha_t = (1-\beta_t), \overline{\alpha_t} = \prod_{t=1}^{T}(1-\beta_t) \tag{3}$$

Where $\varepsilon \sim \mathcal{N}(0, I)$ and correlates with t.

When $T \to \infty$, x_T will converge to standard Gaussian noise $\mathcal{N}(0, I)$.

B. Process of Backward Denoising

The original distribution x_0 can be restored from a standard Gaussian distribution x_T by the inverse transformation $q(x_{T:0})$. However, due to the intractable $q(x_{T:0})$, the process of backward denoising can be parametrized by virtue of an approximation to $q(x_{T:0})$ as:

$$p_\theta(x_{T:0}) \sim q(x_{T:0})$$
$$p_\theta(x_{T:0}) = p(x_T) \prod_{t=1}^{T} p_\theta(x_{t-1} \mid x_t) \tag{4}$$
$$= p(x_T) \prod_{t=1}^{T} \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \delta_\theta(x_t, t))$$

Where $p_\theta(x_{T:0})$ denotes an approximation to $q(x_{T:0})$ and $\mu_\theta(x_t, t), \delta_\theta(x_t, t)$ are the mean and standard deviation of noise ε_θ ,which needs to be removed from $p_\theta(x_{t-1} \mid x_t)$.

With the help of powerful computational capabilities, deep learning model can be applied to estimate ε_θ in process of backward denoising. In formula (5)-(6), KL

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divergence[16] helps measure the difference between two probability distributions and the use of Variable Lower Boundary(VLB)[17] leads to a better approximation of the true posterior distribution.

$$L_t = D_{KL}\left(q(x_{t-1}\mid x_t), p_\theta(x_{t-1}\mid x_t)\right) \tag{5}$$

$$\mathbb{E}_{q(x_0)} p_\theta(x_0) \le L_{VLB} = \mathbb{E}_{q(x_{1:T}\mid x_0)}\left[\log \frac{q(x_{1:T}\mid x_0)}{p_\theta(x_{T:0})}\right] \tag{6}$$

The goal of the backward denoising process is to optimize the log-likelihood of the model's predicted distribution $p_\theta(x_0)$ under the true data distribution $q(x_0)$.

III. Short Term Scenarios Generation Based on Conditional Diffusion Model

Influenced by various uncertainties, wind power scenarios generation requires considering the internal characteristics in temporal dependencies and its relationship with forecast errors. In this paper, we proposed the Conditional Diffusion Model, where conditional guidance term was incorporated as well as the Res-Unet to enhance the expressive power and training stability of the diffusion model for short term scenarios generation.

A. Modeling on Forecast Errors Distribution

Given the continuous growth of wind power installed capacity, the impact of wind power forecast errors cannot be overlooked. Taking into account the influence of wind power forecast errors, it is important to establish a connection between the distribution of forecast errors and the conditional guidance term in diffusion model.

Define $P(e\mid f)$ as joint discrete probability distribution of forecast error e with respect to the wind power forecast value f. The expectation of the forecast error under forecast interval can be calculated by following formula:

$$\begin{aligned} \phi_i &= \sum e \frac{P(e\mid f_{\in D_i})}{P(f_{\in D_i}\mid f)} \\ D_i &= \frac{f_{\max}-f_{\min}}{f_{\max}} \times [\frac{i-1}{n}, \frac{i}{n}) \end{aligned} \tag{7}$$

Where ϕ_i denotes expectation of forecast error under forecast interval D_i, n denotes the number of interval into which forecast value is divided, $P(e\mid f_{\in D_i})$ denotes joint probability distribution of error in forecast interval D_i, $P(f_{\in D_i}\mid f)$ is probability of D_i in all forecast value.

The non-parametric kernel density estimation method is applied to fit the prediction error distribution. Function of the kernel density estimation model is as follows

$$K_h(x) = \frac{1}{nh}\sum_{i=1}^{n} K(\frac{x-\phi_i}{h}) \tag{8}$$

In this equation, $K_h(x)$ represents the function of estimated kernel density of the errors, h is the bandwidth parameter, and K is the kernel function. Kernel density estimation method to fit forecast errors distribution with respect to forecast interval is shown in Fig.2

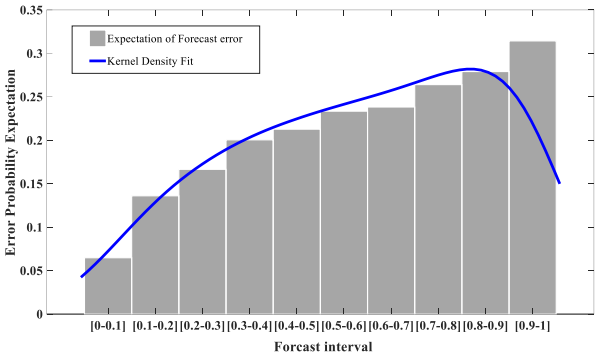

Fig.2 Forecast errors expectation distribution and its kernel density fit function

The conditional guidance term c is an interval formed by formula (10):

$$\begin{aligned} c_{up} &= \min(1, f + K_h(f)) \\ c_{down} &= \max(0, f - K_h(f)) \\ c &= [c_{down}, c_{up}] \end{aligned} \tag{9}$$

Then c is utilized as a guidance term to adjust $p_\theta(x_{t-1}\mid x_t)$, thus formula (4) will be updated as:

$$\begin{aligned} p_\theta(x_{T:0}) &= p(x_T)\prod_{t=1}^{T}\mathcal{N}(x_{t-1};\mu_\theta(x_t,t\mid c),\delta_\theta(x_t,t)) \\ &= p(x_T)\prod_{t=1}^{T}\mathcal{N}(x_{t-1};\mu_\theta(x_t,t)+\nabla_{x_t}\gamma\|x_t-c\|,\delta_\theta(x_t,t)) \end{aligned} \tag{10}$$

Where $\nabla_{x_t}\|\gamma\cdot x_t - c\|$ is a gradient term related to the distance of x_t and c to modify $\mu_\theta(x_t,t)$. γ is a 0-1 binary coefficient that helps x_t avoid getting trapped in the conditional interval.

B. Network Architecture and Scenarios Generation

The conditional diffusion model for wind scenario generation can be divided into three main functional frameworks: diffusion, noising estimation and denoising parts. The framework of conditional diffusion model is shown in Fig.3

The diffusion part is designed for realizing the process of forward diffusion demonstrated in Section Ⅱ by mixing a set of predefined gaussian noise ε with the time embedding $t=\{1,2,3,...,T\}$ into the original data x_0 (usually historical actual wind power output).

The noising estimation part intends to obtain pattern of ε_θ. The Res-UNet structure is adopted in our work which combines the strengths of U-Net and ResNet to handle deeper architectures and enhance scenarios generation performance.

The U-Net architecture consists of two pathways in a

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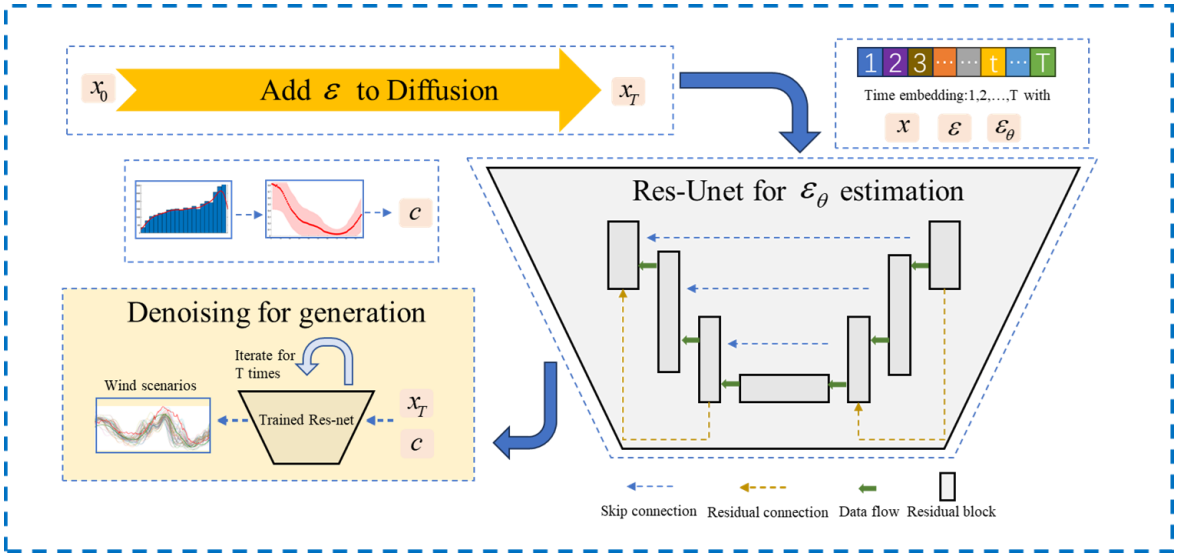


Fig.3 Framework of conditional diffusion model for wind scenario generation

U-shaped manner: down-sample path and up-sample path. The former extracts high dimensional features from input information and gradually compresses them to lower dimension. The latter restores the expected output by spreading the compressed information across hierarchically increasing channels.

The introduction of residual network facilitates U-net to handle deeper architectures and alleviate the vanishing gradient problem. In U-net, sequential connected stacks of residual block serve as fundamental building block that each residual block typically contains multiple 1-D convolutional layers, activation functions and layer normalization. The learned information will flow through layers as well as be propagated directly to next block. Besides, skip channels are used to concatenate feature from the down-sample path to the corresponding up-sample path, enabling the preservation and transfer of detailed information. This helps in the reconstruction and refinement of the final output.

The parameter updated strategy is minimizing Mean Squared Error (MSE) loss between the estimated Gaussian noise ε_θ and the noise in process of forward diffusion ε . The loss function can be represented as follows:

$$L_{model} = \min \mathbb{E}_{t,x_0,\varepsilon}\left[\left\|\varepsilon - \varepsilon_\theta\left(\overline{\alpha}_t x_0 + \sqrt{1-\alpha_t}\,\varepsilon, t\right)\right\|^2\right] \quad (11)$$

Finally, once the loss of model converge, scenarios can be generated through the trained model by denoising ε_θ from $x_T \sim \mathcal{N}(0, I)$ according to formula(10) .

IV. Case Study

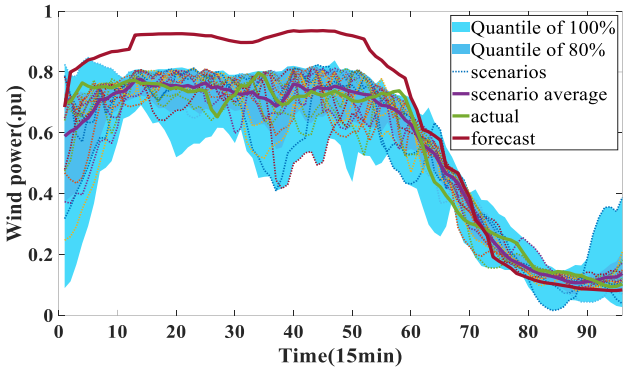
A. Data Description

The dataset comes from a wind farm cluster in the northwest region of China. The dataset includes forecasted and actual power output data from 2021, each with a time resolution of 15 minutes, 35040 data points. The annual time series is reconstructed to sequences of length 96, with a total of 365 curves and the value is normalized to per unit for short term wind scenarios generation. Randomly select 340 curves of actual wind power output data as the training set for model training and the remaining as validation set.

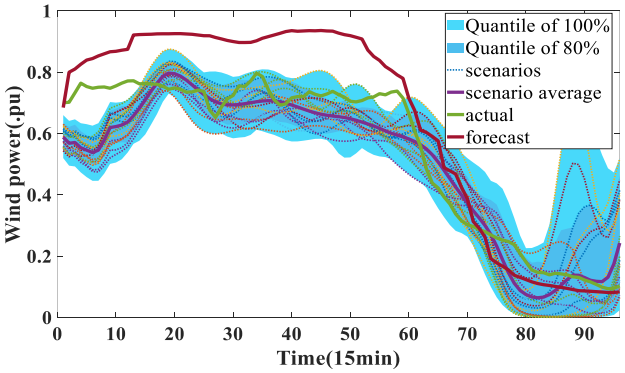
The model was trained on NVIDIA GeForce 3060(6GB RAM), Intel i7-12700H 2.70-GHz CPU with configuration environment of Python 3.9, torch version 1.12.

B. Results Validation

Fig.4 shows the scenarios interval consisted of 50 generated curves within which 20 curves are selected to plot against the corresponding forecasted and actual wind power output curves. The methods of CGAN in [10] is compared with method proposed in this paper.



(a) proposed method



(b) CGAN method

Fig.4 Scenarios generated by proposed method and CGAN

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Both methods can accurately capture the trends in wind power actual output. However, the CGAN method generates scenarios with less diversity, presenting smoothness of curves and lower temporal resolution. The proposed method generates scenarios with rich wind power fluctuation details and higher diversity.

The generated wind power output scenarios need to effectively represent the uncertainty of wind power output. In this study, four metrics, namely energy score, coverage rate, scenario width, and autocorrelation coefficient (ACF), are used to evaluate the rationality and effectiveness of the generated scenarios.

The coverage rate describes the extent to which the generated scenario set envelops the actual wind power output.

$$CR_{\alpha} = \frac{T_{\alpha}}{T} \times 100\% \tag{12}$$

Where T_{α} represents the number of curves when the actual wind power output falls within the α quantile interval of scenarios. T represents the total number of time intervals in the wind power output sequence. CR_{α} represents the coverage rate under α quantile interval. A higher value indicates a higher validity of the generated scenario set.

Scenario width is used to evaluate the random uncertainty in the distribution of generated scenarios. A smaller average width indicates lower uncertainty in the generated scenario set and a more concentrated distribution.

$$\mathrm{PIAW}_{\alpha} = \frac{1}{T}\sum_{n=1}^{T}(P_{n,\alpha}^{up} - P_{n,\alpha}^{down}) \tag{13}$$

Where PIAW_{α} represents the average width of the power interval under α quantile interval. $P_{n,\alpha}^{up}$ represents the upper bound of the wind power output scenario set at time t under α quantile interval. $P_{n,\alpha}^{down}$ represents the lower bound of the wind power output scenario set at time t under α quantile interval.

Energy Score[18] is used to assess probabilistic forecasts of a multivariate quantity. It provides a measure of the quality or accuracy of the generated scenarios compared to reference data:

$$ES = \frac{1}{T}\sum_{j}^{T}\left\|S_j - Y\right\| - \frac{1}{2T^2}\sum_{i}^{T}\sum_{j}^{T}\left\|S_j - S_i\right\| \tag{14}$$

Where S_i is the members of generated scenarios, Y is the actual wind power output, $\|\cdot\|$ denotes the Euclidean norm. A lower energy score indicates better performance, meaning the generated scenarios align closely with the reference data.

Scenarios set of the same day generated by method proposed in this paper and CGAN method in [10] are compared in Table 1 set with 100% and 90% quantiles:

Table 1 Metrics evaluation

	Quantile	Our Work	CGAN
Energy	100%	0.3242	0.6316
Score	80%	0.3620	0.6918
Scenario	100%	26.24%	26.47%
Width	80%	12.42%	15.62%
Coverage	100%	96.88%	79.71%
Rate	80%	73.96%	60.42%

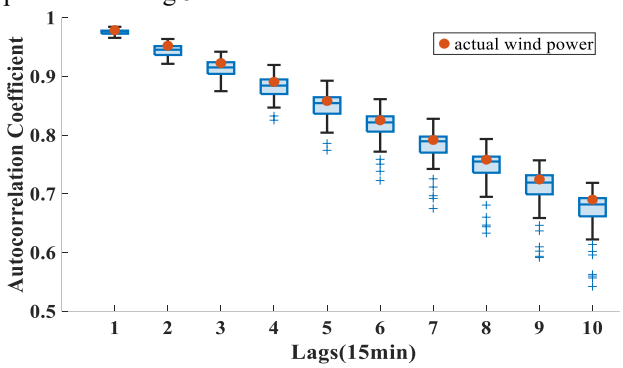
The method proposed in this paper outperforms the CGAN method, exhibiting a lower energy score. It demonstrates better coverage rate in both 100% and 90% quantile interval with narrower interval widths.

The ACF is calculated by taking the correlation coefficient between the original time series and a lagged version of itself as:

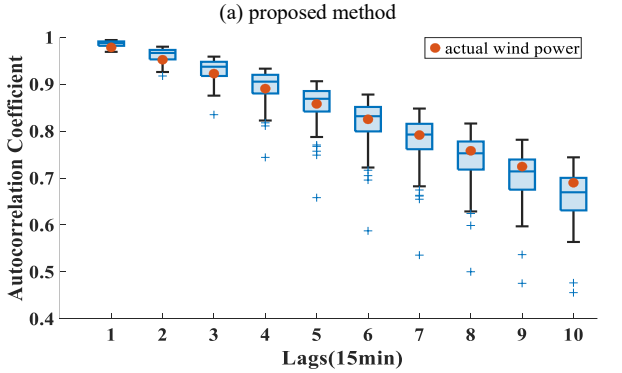
$$\hat{\rho}_k = \frac{\sum_{t=1}^{n-k}\left(P_t - \bar{P}\right)\left(p_{t+k} - \bar{P}\right)}{\sum_{t=1}^{n}\left(P_t - \bar{P}\right)^2} \tag{12}$$

Where k represents the time lags, $\bar{P}$ represents the average wind power, $\hat{\rho}_k$ represents the autocorrelation coefficient of wind power output with a time lags of k. The coefficient $\hat{\rho}_k$ decreases gradually from 1 to 0 as the lag time increases, where a value of 1 indicates a perfect positive correlation, and 0 indicates no correlation. Therefore, the quality of the generation scenario can be assessed by analyzing the magnitude and changing characteristics of the correlation coefficient.

ACF distributions of scenarios in Fig.4 are compared with the ACF values of the actual wind power output and presented in Fig.5.



(a) proposed method



(b) CGAN method

Fig.5 ACF distribution of generated scenarios and ACF of actual wind power output

The correlation of the actual power output is well

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captured by both generated scenarios, and the trend of changes in the ACF of the generated scenarios aligns with the trend of changes in ACF of the actual wind power time series. In contrast to CGAN method, the proposed approach in this paper generates scenarios with a more concentrated ACF distribution, and the magnitude of deviated outliers is smaller.

V. Conclusion

Diffusion model is a powerful framework that has gained significant attention for generative tasks. In this paper, we present an in-depth exploration of a novel deep learning method for short-term wind power scenario generation based on conditional diffusion model by discussing its underlying theory, the incorporation of conditional term and design of model architecture. Assessed by case study and comparison with CGAN method, conditional diffusion model is proved to be a robust and flexible approach for wind scenario generation that can effectively capture the inherent uncertainty and variability of wind power.

In future, we intend to extend our work to diverse renewable energy sources such as PV. In addition, a modification to model architecture can be tailored for various temporal scenario generation.

References

[1] International Renewable Energy Agency. (2023). Renewable Energy Statistics [DB/OL]. Abu Dhabi, United Arab Emirates: International Renewable Energy Agency. Accessed July 5 2023, https://www.irena.org/Data/View-data-by-topic/Capacity-and-Generation/Regional-Trends

[2] XI Jinping. Speech at the general debate of the 75th session of the United Nations General Assembly[N]. Gazette of the State Council of the People's Republic of China，2020(28):5-7(in Chinese).

[3] National Energy Administration. Blueprint for the Development of New Power Systems [EB/OL]. Beijing. NEA , 2023(2023-06-02). http://www.nea.gov.cn/2023-06/02/c_1310724249.htm

[4] LI Pai，LIU Chun，HUANG Yuehui，et al. Modeling correlated power time series of multiple wind farms based on hidden Markov model [J]. Proceedings of the CSEE，2019，39(19)：5683-5691，5896(in Chinese).

[5] A. Sturt and G. Strbac, "Efficient Stochastic Scheduling for Simulation of Wind-Integrated Power Systems," IEEE Transactions on Power Systems, vol. 27, no. 1, pp. 323-334, 2012.

[6] P. Chen, T. Pedersen, B. Bak-Jensen, and Z. Chen, "ARIMA-Based Time Series Model of Stochastic Wind Power Generation," IEEE Transactions on Power Systems, vol. 25, no. 2, pp. 667-676, 2010.

[7] R. Zarate-Minano, M. Anghel, and F. Milano, "Continuous wind speed models based on stochastic differential equations," APPLIED ENERGY, vol. 104, pp. 42-49, APR 2013.

[8] M. Arjovsky, S. Chintala, and L. Bottou, "Wasserstein Generative Adversarial Networks," presented at the INTERNATIONAL CONFERENCE ON MACHINE LEARNING, VOL 70, 2017.

[9] M. Mirza and S. Osindero, "Conditional generative adversarial nets," arXiv preprint arXiv:1411.1784, 2014.

[10] HUANG Yuehui，SUN Yanan，LI Chi，Song Ziyuan. Generation method of short-term output scenarios for multi-regional wind power based on conditional generative adversarial network[J] . Power System Technology，2023，47(1)：63-77(in Chinese).

[11] R. Yuan, B. Wang, Y. Sun, X. Song, and J. Watada, "Conditional Style-Based Generative Adversarial Networks for Renewable Scenario Generation," IEEE Transactions on Power Systems, vol. 38, no. 2, pp. 1281-1296, 2023.

[12] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," Advances in neural information processing systems, vol. 33, pp. 6840-6851, 2020.

[13] Z. Kong, W. Ping, J. Huang, K. Zhao, and B. Catanzaro, "Diffwave: A versatile diffusion model for audio synthesis," arXiv preprint arXiv:2009.09761, 2020.

[14] N. Ruiz, Y. Li, V. Jampani, Y. Pritch, M. Rubinstein, and K. Aberman, "Dreambooth: Fine tuning text-to-image diffusion models for subject-driven generation," in Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, 2023, pp. 22500-22510.

[15] F. Mazé and F. Ahmed, "Diffusion models beat GANs on topology optimization," in Proceedings of the AAAI Conference on Artificial Intelligence (AAAI), Washington, DC, 2023.

[16] J. R. Hershey and P. A. Olsen, "Approximating the Kullback Leibler divergence between Gaussian mixture models," in 2007 IEEE International Conference on Acoustics, Speech and Signal Processing-ICASSP'07, 2007, vol. 4: IEEE, pp. IV-317-IV-320.

[17] D. P. Kingma and M. Welling, "Auto-encoding variational bayes," arXiv preprint arXiv:1312.6114, 2013.

[18] T. Gneiting, L. I. Stanberry, E. P. Grimit, L. Held, and N. A. Johnson, "Assessing probabilistic forecasts of multivariate quantities, with an application to ensemble predictions of surface winds," TEST, vol. 17, no. 2, pp. 211-235, 2008/08/01 2008.

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